File Copy Studio

User Guide · Version 2026.09.08.0002

A WPF folder-to-folder copy/move utility with configurable conflict resolution, CRC verification, crash-safe resume, and built-in job scheduler.

Version `2026.09.08.0002` -- .NET 10 · C# 14 · WPF · MVVM · JSON SourceGen

Features

Feature	Detail
Conflict resolution	Overwrite, Overwrite-if-newer, Overwrite-if-older, Skip-if-same-date, Skip-all, Rename
CRC32 verification	Post-copy verify; standalone Verify Only mode; `.crc` sidecar option
Crash-safe resume	`.fctemp` + `.fctemp.meta` sidecar; survives power loss; auto-detected on next scan
Parallel workers	Auto-detects same vs. different volumes; configurable 1-8× threads
Preserve timestamps & attributes	Optional; respects NTFS extended attributes
Delete source (Move)	Confirmation prompt; safe -- deletes only after verified copy
Dry run	Plans actions, shows file grid, writes nothing to disk
Throttle	Bandwidth cap in Mbps; per-session or global pref
Symlink policy	Follow, Skip, or Copy-as-link
File masks	Include/exclude semicolon-separated wildcards (`*.jpg;*.png`)
Job scheduler	Built-in cron-style scheduler; runs jobs unattended; per-job throttle/conflict options
5 named presets	Slot 1-5; persist paths + all options; one-click reload; Ctrl+1-5
Save/load job files	`.fclist` JSON format; repeatable batch transfers
Shell context menu	Right-click folder/drive -> "Copy with File Copy Studio"; auto-scans
Portable mode	`portable.flag` next to exe; prefs + logs in `Data\`; no registry; USB-ready
Tray minimize	Optional; balloon notification + system sound on completion
Pause on battery	Automatically pauses when on battery; resumes on AC
WCAG 2.2 AAA	7:1 contrast ratios, 44px touch targets, 2px Gold/Cyan focus rings, live regions, glyph+color status
Light / Dark / Auto themes	Auto follows Windows accent; runtime switch; high-contrast passthrough

Feature	Detail
Single instance	Second launch forwards args to running instance via named pipe

UI Layout

```
+-------------------------------------------------------------+
| 1. Identity / Scale File Copy Studio [Standard 1×]|
+-------------------------------------------------------------+
| 2. Nav row |
| [ Source] [ Dest] [ Swap] [ Scan] [v Verify] |
| [ Start] [ Pause] [x Cancel] [ Dry Run] |
| [ Load Job] [ Save Job] [ Scheduler] [ About] |
+-------------------------------------------------------------+
| 3. Path / Progress |
| Source _____________________________ [Browse] |
| Dest _____________________________ [Browse] |
| ###############..... 47% [filename.ext] |
| ####.............. 23% total 1.2 GB/s |
+-------------------------------------------------------------+
| 4. Options / Sliders |
| On conflict: [Overwrite if newer v] |
| [x] Subfolders [x] Timestamps [x] Attributes [x] Verify CRC |
| [ ] Move [x] Resume [x] Auto-parallel |
| [ ] Tray [x] Sound [ ] Pause on battery |
| Buffer ============== 1024 KB Parallel == 2× |
| Throttle == 0 Mbps (unlimited) |
| Include [*.*] Exclude [] Symlinks [Follow v] |
| 12,847 files -- Copy 8,201 · Overwrite 412 · Skip 4,234 |
+-------------------------------------------------------------+
| 5. File grid (sortable, filterable, resizable columns) |
| Filter: [All v] ! + (reload) ... | File | Size | Modified |
| ------------------------------------------------- |
| Error log ! 2 errors [ Copy errors] [ Clear] |
+-------------------------------------------------------------+
| 6. Presets [*1] [2] [3] [*4] [5] [ Save Preset] |
+-------------------------------------------------------------+
 (resize grip)
```

Keyboard Shortcuts

Keys	Action
`Ctrl+S`	Scan
`F5`	Start copy
`Esc`	Cancel
`Ctrl+P`	Pause / Resume
`Ctrl+D`	Dry run
`Ctrl+V` (main window)	Verify only
`Ctrl+1` - `Ctrl+5`	Load preset slot
`Ctrl+Shift+1` - `Ctrl+Shift+5`	Save to preset slot

File Formats

`.fclist` *-- Job file*

JSON file saving paths + options + planned actions. Load with `Load Job`, save with `Save Job`. Clicking Scan after load refreshes planned actions against current files.

```
{
 "FormatVersion": "1",
 "CreatedUtc": "2026-05-09T12:00:00Z",
 "SourcePath": "D:\\Photos",
 "DestPath": "E:\\Backup\\Photos",
 "ConflictMode": "OverwriteIfNewer",
 "IncludeSubdirs": true,
 "VerifyCrc": true,
 "Items": [
 { "RelativePath": "2024\\img_001.jpg", "SizeBytes": 4194304, "PlannedAction": "Copy" }
 ]
}
```

`.fctemp` + `.fctemp.meta` *-- Resume markers*

Written alongside the destination during a copy. If interrupted, the next scan detects them and resumes from the last 64 MB checkpoint.

Do not delete `.fctemp` **files manually** unless you want to start fresh -- use right-click -> `Discard partial` in the file grid context menu instead.

```
dest\large_video.mkv.fctemp <- partial payload (raw bytes)
dest\large_video.mkv.fctemp.meta <- JSON sidecar:
 { "SourceLength": 10737418240, "SourceMtimeUtcTicks": ...,
 "PartialCrc": 3735928559, "PartialOffset": 1073741824 }
```

If the source file changes between runs, the partial is discarded automatically and the copy restarts.

`.crc` *-- CRC sidecar (optional)*

When **Write CRC sidecar** is enabled, a `.crc` file is written alongside each destination file containing the CRC32 hash. Subsequent verify runs use these sidecars for fast integrity checks without re-reading the source.

Scheduler

Open the scheduler with `Scheduler` or `Ctrl+Shift+S`. Jobs run in the background while the main window is visible or minimised to tray.

Each job stores its own source/dest paths, conflict mode, include/exclude masks, throttle, and schedule expression. Jobs can be enabled/disabled individually without deleting them.

Schedule expressions use standard 5-field cron syntax:

```
+--------- minute (0-59)
| +------- hour (0-23)
| | +----- day of month (1-31)
| | | +--- month (1-12)
| | | | +- day of week (0-7, Sun=0/7)
| | | | |
* * * * *
```

Examples: `0 2 * * *` = daily at 02:00 · `0 * * * *` = hourly · `30 18 * * 5` = Fridays at 18:30

CLI / Shell Integration

```
FileCopyStudio.exe [--source <path>] [--dest <path>] [--scan]
FileCopyStudio.exe D:\MyFolder # positional path treated as --source
```

The installer registers shell verbs for `Folder`, `Directory`, `Directory\Background`, and `Drive` -- all pass `--source "%1" --scan` automatically.

If File Copy Studio is already running, the new instance forwards its arguments over a named pipe and exits; the running window activates and applies the args.

Build

Prerequisites: .NET 10 SDK, NSIS 3.x (optional, for installer)

```
cd FileCopy
build.bat
```

Auto-increments version (`yyyy.MM.dd.NNNN`), patches `FileCopy.csproj` and `installer.nsi`, compiles a self-contained Release single-file exe, then calls `makensis installer.nsi` if NSIS is on PATH.

Portable / USB build

```
cd FileCopy
build-portable.bat
```

Produces `FileCopyStudio_Portable_<version>.zip`:

```
FileCopyStudio.exe <- self-contained, no .NET runtime needed
portable.flag <- activates portable mode (delete to use %APPDATA%)
manual.pdf <- user manual
Data\
 prefs.json
 logs\
```

Portable mode: prefs + logs stored in `Data\` next to the exe; no registry writes; title bar shows `[Portable]`.

Tests

```
cd FileCopy.Tests
test.bat
```

Covers `PartialFile`, `DriveAffinity`, `CopyEngine.PlanActions`, `FcJob`, `SizeFormat`, and `AppPrefs`.

Stack

`.NET 10` · `WPF` · `C# 14 (field keyword)` · `MVVM (MainWindowViewModel)` · `JSON Source Generation` · `System.IO.Hashing (CRC32)` · `System.IO.Pipes.AccessControl` · `Microsoft.Extensions.Logging`